Università Commerciale Luigi Bocconi

20532 Macroeconometrics

Problem Set 2

Explanatory Note

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Introduction and AI Disclosure

This explanatory note presents and discusses the numerical exercises contained in Problem Set 2. The analysis focuses on unit-root testing, spurious regressions, invertibility and structural impulse responses, and Granger causality in a multivariate time-series setting.

The MATLAB implementation is contained in `PS2_complete_solution.m`, together with any auxiliary functions required by the analysis.

Finally, regarding the AI disclosure, we used Codex and ChatGPT (Sol High Model) to polish the text and layout of the explanatory note and to improve the readability and portability of the MATLAB code.

Question 1: Unit Root Testing

Throughout Question 1 we simulate the pure random walk

$$y_t = y_{t-1} + \varepsilon_t, \qquad \varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1),$$

with $y_0 = 0$. We choose the innovation variance $\sigma_\varepsilon^2 = 1$, generate $T = 250$ observations in each replication, and repeat the experiment $N = 10{,}000$ times.

The course notes define a random walk as

$$y_t = y_{t-1} + \varepsilon_t,$$

or, equivalently, as an AR(1) process with autoregressive coefficient $\phi = 1$. Iterating the process backwards gives

$$y_t = y_0 + \sum_{j=1}^{t} \varepsilon_j.$$

Consequently,

$$\mathbb{E}[y_t] = y_0, \qquad \text{Var}(y_t) = t\sigma_\varepsilon^2.$$

Since the variance depends on time and grows without bound, the random walk is non-stationary. It is a unit-root, or difference-stationary, process: taking first differences gives

$$\Delta y_t = \varepsilon_t,$$

which is stationary.

Question 1(a): Empirical Distribution of the OLS Estimator

> *Compute the empirical distribution of the OLS estimator in the case of a pure random walk (no drift) with $T = 250$; you are free to choose the variance of the innovation.*

We simulate the random walk above in every Monte Carlo replication and estimate its autoregressive coefficient by OLS.

According to the course notes, the usual stationary AR(1) asymptotic argument cannot be applied when $\phi = 1$ so that standard normal inference cannot be evoked. In the stationary case, $\sqrt{T}(\widehat{\phi} - \phi)$ has a standard asymptotic treatment, but this derivation relies on $1 - \phi^2 \neq 0$. At the unit root, $\phi = 1$, this condition fails.

In particular, the unit–root OLS estimator is *super-consistent*:

$$\sqrt{T}(\widehat{\phi} - 1) \xrightarrow{p} 0,$$

so that a stronger scaling (T instead of $\sqrt{T}$),

$$T(\widehat{\phi} - 1),$$

is needed to obtain a non-degenerate limiting distribution, that is, a functional of Brownian motion (of course a non-Normal limiting distribution): on this point, see Ghysels and Marcellino, *Applied Economic Forecasting Using Time Series Methods*, Chapter 5, Section 5.5, around pp. 191–192. In summary, the sampling distribution is non-standard and therefore the usual t/F test rules do not apply.

Our MATLAB implementation reports both the OLS slope from the no-intercept regression and, for comparison, the slope obtained when an intercept is included. The latter is useful for comparison with the Dickey–Fuller specification employed in the following parts.

Table 1: Empirical distribution of the OLS autoregressive coefficient under the pure random walk. Results are based on 10,000 Monte Carlo replications.

Estimator	Mean	SD	Median	2.5%	97.5%
No intercept	0.99295	0.01266	0.99665	0.95860	1.00670
With intercept	0.97865	0.01794	0.98280	0.93268	1.00180

The left panel of Figure 1 displays the two empirical distributions. Both are concentrated close to the unit-root value $\phi = 1$. For the no-intercept specification, the Monte Carlo mean is 0.99295 and the median is 0.99665. With an intercept, the corresponding mean is 0.97865 and the distribution is more dispersed. Notably, even if the true value is $\phi = 1$, the OLS estimator proves having a small downward bias as it has a smaller than one mean in finite samples.

The important theoretical point is that, at $\phi = 1$, the standard stationary AR(1) asymptotic distribution does not apply.

Question 1(b): Dickey–Fuller t-Test

> *Construct a t-test for the null hypothesis $H_0 : \rho = \phi - 1 = 0$ in the test regression*
>
> $$\Delta y_t = \alpha + \rho y_{t-1} + \varepsilon_t,$$
>
> *against the one-sided alternative $H_1 : \rho < 0$. Using a standard Normal distribution, how often do you reject the null hypothesis at the 95% confidence level? Is the actual distribution of the t-test symmetric? Discuss.*

The Dickey–Fuller parametrization used in the notes is

$$\Delta y_t = \alpha + \rho y_{t-1} + \varepsilon_t, \qquad \rho = \phi - 1.$$

Therefore,

$$\rho = 0 \quad \Longleftrightarrow \quad \phi = 1,$$

so testing

$$H_0 : \rho = 0 \qquad \text{against} \qquad H_1 : \rho < 0$$

is a unit-root test.

For each replication, we estimate the regression above and construct the usual t-ratio for $\widehat{\rho}$,

$$t_{\widehat{\rho}} = \frac{\widehat{\rho}}{\widehat{\text{se}}(\widehat{\rho})}.$$

The course notes stress that, under the unit-root null, the Central Limit Theorem does not generate the usual Standard Normal distribution. The limiting distribution is instead the non-standard Dickey–Fuller distribution. The notes also stress that this distribution is asymmetric, so the usual Normal critical values cannot be used. This is therefore what we expect to find.

If we nevertheless use the one-sided Standard Normal 5% critical value -1.645, our Monte Carlo experiment rejects the true unit-root null in 45.96% of the replications.

This is exactly the type of incorrect inference highlighted in the course notes: using standard Normal critical values in the unit-root case leads to excessive rejection of the null.

In our simulation, the empirical Dickey–Fuller t distribution has

$$\text{mean}(t_{\widehat{\rho}}) = -1.5331, \qquad \text{median}(t_{\widehat{\rho}}) = -1.5635.$$

Its empirical skewness is 0.1618, so the finite-sample distribution is not exactly symmetric. The main point, however, is that the distribution is clearly not $\mathcal{N}(0,1)$, as can be seen in the central panel of Figure 1.

Table 2: Rejection frequencies of the unit-root null using the conventional Normal and Dickey–Fuller critical values.

Critical-value distribution	5% critical value	Rejection frequency
Standard Normal	−1.6449	45.96%
Dickey–Fuller	−2.8800	4.88%

To sum up, the Standard Normal critical value is inappropriate under a unit root. Using −1.6449 leads to rejection of the true null in **45.96**% of the simulations. The empirical test-statistic distribution is not Standard Normal and is not exactly symmetric, consistent with the non-standard Dickey–Fuller framework discussed in the notes.

Question 1(c): Empirical Dickey–Fuller Percentiles

> *Compute some percentiles of the empirical distribution of the t-test generated in point (b) and check that they are close to those tabulated by Dickey and Fuller.*

The course notes state that the unit-root t-statistic must be compared with the specific critical values tabulated by Dickey and Fuller rather than with Standard Normal critical values. They also emphasize that the appropriate distribution depends on the deterministic components included in the test regression, such as an intercept or a time trend.

Since our Dickey–Fuller regression contains an intercept but no deterministic trend, we compare the simulated percentiles with the corresponding Dickey–Fuller table used in the MATLAB code (see pag.763 of Hamilton, *Time Series Analysis*).

Table 3: Empirical and tabulated Dickey–Fuller percentiles for $T = 250$ with an intercept and no deterministic trend.

Percentile	Empirical	DF table	Difference
1%	−3.5090	−3.4600	−0.0490
2.5%	−3.1708	−3.1400	−0.0308
5%	−2.8716	−2.8800	0.0084
10%	−2.5663	−2.5700	0.0037
90%	−0.4288	−0.4200	−0.0088
95%	−0.0597	−0.0600	0.0003
97.5%	0.2463	0.2400	0.0063
99%	0.5819	0.6200	−0.0381

The empirical percentiles are close to the tabulated Dickey–Fuller values. In particular, at the 5% level we obtain -2.8716 from the Monte Carlo distribution, compared with the tabulated value -2.88. Small discrepancies are acceptable because of both Monte Carlo error and the finite $T = 250$.

Using the Dickey–Fuller critical value -2.88, the rejection frequency is 4.88%, which is very close to the nominal 5% level.

This contrasts sharply with the 45.96% rejection frequency obtained when the Standard Normal critical value is used. The simulation therefore reproduces the main lesson of the notes: unit-root inference requires Dickey–Fuller critical values because the usual Normal distribution does not apply!

Question 1(d): Joint Unit-Root Test

> *Construct an F-test for the null hypothesis*
>
> $$H_0 : \alpha = 0, \qquad \rho = 0,$$
>
> *against the alternative that there is no unit root, first using a χ^2 distribution (how many degrees of freedom?) and then the true distribution of the test. How often do you reject H_0 at the 95% confidence level in both cases?*

The course notes stress that the deterministic terms included in a unit-root test are part of the hypothesis being tested. With the regression

$$\Delta y_t = \alpha + \rho y_{t-1} + \varepsilon_t,$$

setting only $\rho = 0$ gives

$$\Delta y_t = \alpha + \varepsilon_t,$$

which is a random walk with drift. For a non-trending random walk with no drift, the strict null is therefore the joint restriction

$$H_0 : \qquad \rho = 0 \quad \text{and} \quad \alpha = 0.$$

This is the joint hypothesis described in the notes, where the corresponding test is referred to as the Dickey–Fuller Φ_1 test.

There are two restrictions under the null. The conventional Wald comparison therefore uses a χ^2 distribution with two degrees of freedom ($\chi^2(2)$); for the general asymptotic result that, when testing q restrictions, $qF \overset{a}{\sim} \chi^2(q)$, see Ghysels and Marcellino, *Applied Economic Forecasting Using Time Series Methods*, Chapter 3, Section 3.3, around p. 126.

In the MATLAB implementation, the Wald statistic W and the corresponding F statistic satisfy $W = 2F$.

The 5% conventional $\chi^2(2)$ critical value is 5.9915.

Using this conventional reference distribution, the true joint null is rejected in 19.41% of the Monte Carlo replications.

The course notes emphasize, however, that the distribution of the unit-root F statistic is non-standard. For the present specification, the appropriate Dickey–Fuller Φ_1 critical value used in the code is 4.63 (see Hamilton, *Time Series Analysis*, Table B.7, p. 764, Case 2, $T = 250$).

Using this critical value, the rejection frequency becomes 5.08%.

Table 4: Joint test of $H_0 : \alpha = 0, \rho = 0$ under the true random-walk DGP.

Reference distribution	5% critical value	Rejection frequency
Conventional $\chi^2(2)$ for W	5.9915	19.41%
Dickey–Fuller Φ_1 for F	4.6300	5.08%

The difference between the two rejection frequencies again reflects the non-standard nature of inference under a unit root. The ordinary reference distribution rejects the true null too frequently, whereas the Dickey–Fuller Φ_1 critical value delivers a rejection frequency very close to the nominal 5% level.

To sum up, the null contains two restrictions, $\alpha = 0$ and $\rho = 0$. Using the conventional $\chi^2(2)$ reference distribution gives a rejection frequency of **19.41**%. Using the appropriate Dickey–Fuller Φ_1 critical value instead gives **5.08**%, close to the nominal 5% level.

In a nutshell

Question 1 illustrates the special inferential properties of a unit-root process. A random walk is non-stationary because its variance grows with time, and the usual stationary AR(1) asymptotic results do not apply when $\phi = 1$.

The Dickey–Fuller test statistics therefore have non-standard distributions. In our Monte Carlo experiment, using the Standard Normal critical value for the t-test leads to a rejection

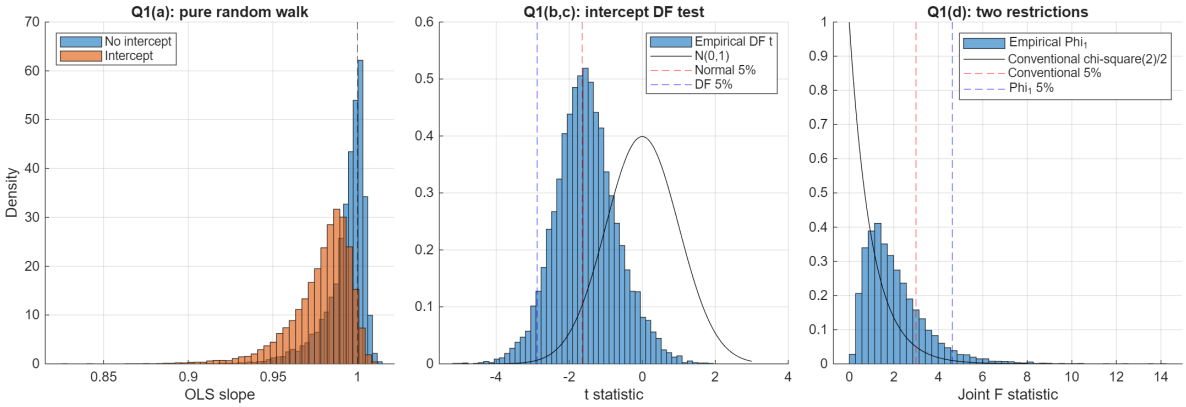


Figure 1: Monte Carlo results for Question 1, based on 10,000 replications with $T = 250$. The left panel reports the empirical distributions of the OLS autoregressive coefficient. The central panel compares the empirical Dickey–Fuller t distribution with the Standard Normal density and reports the Normal and Dickey–Fuller 5% critical values. The right panel displays the empirical distribution of the joint F statistic and the conventional and Dickey–Fuller Φ_1 5% critical values.

frequency of 45.96%, while the appropriate Dickey–Fuller critical value gives 4.88%. Similarly, the conventional $\chi^2(2)$ reference distribution for the joint test gives a rejection frequency of 19.41%, whereas the Dickey–Fuller Φ_1 critical value gives 5.08%.

The simulations therefore reproduce the main conclusion of the course notes: when testing for a unit root, standard asymptotic critical values cannot be used; the relevant Dickey–Fuller non-standard distributions must be employed.

Question 2: Spurious Regression

> *Design a Monte Carlo experiment to show that the regression coefficient, the t-test, and R^2 are meaningless in the case of a spurious regression. In particular, show what happens in each of the four cases discussed in Enders, 3rd or 4th edition, pp. 195–199.*

Simulation and estimation

We investigate when a regression in levels can suggest a relationship between unrelated time series, and how this differs from a genuine cointegrating relationship. For each of four data-generating processes, we run $N = 5{,}000$ replications at each sample size $T \in \{100, 500, 1000\}$. Comparing sample sizes helps distinguish sampling variability that disappears with more observations from a failure of the usual regression assumptions.

The command `rng(2053202,'twister')` fixes the seed once before the loops. Each replication uses fresh innovations generated by `randn`. Throughout, u_t, v_t, and, where needed, η_t are mutually independent sequences of independent $N(0,1)$ random variables. All random walks start at zero and have no drift. Their paths are generated with `cumsum`; no burn-in is applied to them. The stationary AR(1) in case 2 alone has a burn-in of 500 observations.

In every replication we estimate the same regression,

$$y_t = a + bz_t + e_t, \qquad t = 1, \ldots, T.$$

The code forms `X = [ones(T2,1),z]` and computes the intercept and slope with `b = X\y` , MATLAB's backslash operator. Thus all T observations enter the regression, and a constant is included even though the generating processes have zero means at each date. With $\widehat{\theta} = (\widehat{a}, \widehat{b})'$ and $\widehat{e} = y - X\widehat{\theta}$, the conventional OLS calculations are

$$\widehat{\theta} = (X'X)^{-1}X'y, \qquad \widehat{\sigma}^2 = \frac{\widehat{e}'\widehat{e}}{T-2}, \qquad \widehat{V} = \widehat{\sigma}^2(X'X)^{-1}.$$

The script evaluates the inverse factor by `(X'*X)\eye (2)` and obtains the slope standard error as $\widehat{\mathrm{se}}(\widehat{b}) = \sqrt{\widehat{V}_{22}}$. These are the usual homoskedastic OLS standard errors; their validity is one of the properties being examined.

Regression diagnostics

For each simulated sample, we store the slope, the conventional t-statistics, and the coefficient of determination:

$$t(b_0) = \frac{\widehat{b} - b_0}{\widehat{\mathrm{se}}(\widehat{b})}, \qquad R^2 = 1 - \frac{\widehat{e}'\widehat{e}}{\sum_{t=1}^{T}(y_t - \overline{y})^2}.$$

The two-sided test rejects $H_0 : b = b_0$ when $|t(b_0)| > t_{0.975,T-2}$, with the critical value obtained from `tinv(1-alpha/2,T2-2)` and `alpha = 0.05`. The rejection frequency is the fraction of the 5,000 replications satisfying this rule, computed with `mean`. We distinguish $t(0)$ from the statistic evaluated at the true slope: $b^\star = 0$ in cases 1–3 and $b^\star = 1$ in case 4. This distinction separates false rejections of a true null from power against a false null.

Empirical results and the four cases

Table 5 reports the saved results, using `mean` and `std` across replications. Sampling percentiles use `prctile`. The summary is stored in `question2_spurious_summary.csv`.

Table 5: Monte Carlo results, with 5,000 replications per case and sample size.

Case	T	Mean $\widehat{b}$	SD$(\widehat{b})$	Mean R^2	Reject $b = 0$	Reject $b = b^\star$
1	100	−0.0002	0.1018	0.0102	5.04%	5.04%
1	500	−0.0003	0.0450	0.0020	4.94%	4.94%
1	1000	0.0003	0.0314	0.0010	4.78%	4.78%
2	100	−0.0043	0.6626	0.0450	36.62%	36.62%
2	500	−0.0058	0.6874	0.0108	40.60%	40.60%
2	1000	0.0039	0.6930	0.0055	40.22%	40.22%
3	100	0.0114	0.6276	0.2374	76.16%	76.16%
3	500	−0.0015	0.6368	0.2353	89.06%	89.06%
3	1000	−0.0054	0.6237	0.2416	93.12%	93.12%
4	100	0.9101	0.0728	0.8307	100.00%	51.34%
4	500	0.9793	0.0180	0.9593	100.00%	54.26%
4	1000	0.9895	0.0093	0.9791	100.00%	54.74%

SD is the empirical standard deviation across replications. Both rejection columns use conventional two-sided tests at the nominal 5% level. The true slope is $b^\star = 0$ in cases 1–3 and $b^\star = 1$ in case 4; therefore rejection of $b = 0$ in case 4 measures power, while the final column measures rejection under the true null in every case.

Case 1: both series are stationary

We generate $y_t = u_t$ and $z_t = v_t$ with two separate calls to `randn(T2,1)`. Both series are independent $I(0)$ white noise, so $b^\star = 0$ and the classical regression assumptions hold.

As T rises from 100 to 1000, the slope's standard deviation falls from 0.1018 to 0.0314, consistent with the $T^{-1/2}$ rate. Its mean stays near zero, mean R^2 falls from 0.0102 to 0.0010, and rejection rates remain near 5%. The first row of Figure 2 shows the concentrated slope distribution, a t distribution centered near zero, and negligible fit. This validity relies on the white-noise benchmark: stationarity alone does not guarantee valid conventional standard errors with serial dependence.

Case 2: an $I(1)$ dependent variable and an $I(0)$ regressor

The second design is

$$y_t = y_{t-1} + u_t, \qquad z_t = 0.7z_{t-1} + v_t.$$

The code uses `cumsum(randn(T2,1))` for y_t and `filter(1,[1 -0.7],...)` for z_t, retaining the last T of its $T + 500$ observations. This makes initialization effects negligible for z_t. A stationary regressor cannot remove the trend in y_t: for any fixed a, b, $y_t - a - bz_t$ remains nonstationary.

The mean slope stays near zero, but its standard deviation remains around 0.66–0.69: the estimates do not concentrate at zero as T grows. Mean R^2 declines to 0.0055 at $T = 1000$, yet the test rejects the true zero slope in 40.22% of replications. The t-statistic has excessive dispersion (SD 2.3653 at $T = 1000$), but does not diverge as in case 3. This distortion depends on the stationary regressor's persistence in this design. A small R^2 therefore does not make the conventional test reliable.

Case 3: two independent $I(1)$ series

We now generate two independent random walks,

$$y_t = y_{t-1} + u_t, \qquad z_t = z_{t-1} + v_t,$$

using separate calls to `cumsum(randn(T2,1))`. There is no common innovation and no underlying relationship. In particular, no fixed slope can eliminate both independent stochastic trends: at a candidate slope b, the increment of $y_t - bz_t$ is $u_t - bv_t$, with variance $1 + b^2 > 0$.

This is the standard spurious-regression case. The mean slope is again close to zero across replications, but its standard deviation remains about 0.63 at all three sample sizes. At $T = 500$, the central 95% sampling interval is $[-1.3098, 1.2939]$, despite the series being unrelated. A Monte Carlo mean near zero is therefore not evidence of consistency: individual estimates remain widely dispersed as the sample grows.

At the same time, the conventional standard error shrinks even though the slope does not concentrate at zero. The t-statistics consequently grow in magnitude, with empirical standard deviations 7.2297, 16.4138, and 23.5350 at $T = 100, 500, 1000$, respectively. This accords with their $\sqrt{T}$ order of divergence. The nominal 5% test rejects in 76.16%, 89.06%, and 93.12% of replications: adding observations makes the false evidence of a relationship more frequent.

Mean R^2 remains around 0.24 instead of approaching zero; its 97.5th percentile at $T = 500$ is 0.7531. Thus unrelated random walks can display substantial apparent fit. The third row of Figure 2 shows the broad slope distribution, very wide t distribution, and dispersed R^2. The latter has a nondegenerate limiting distribution: it need not approach one, but it does not vanish as in the independent stationary case. The coefficient, conventional significance, and apparent fit cannot be interpreted as evidence of an economic relationship.

Case 4: two cointegrated $I(1)$ series

Finally, we generate a common random walk and add independent transitory noise:

$$s_t = s_{t-1} + \eta_t, \qquad y_t = s_t + u_t, \qquad z_t = s_t + v_t.$$

In MATLAB, `common_trend = cumsum(randn(T2,1))` supplies the shared component, and separate `randn` calls supply the two noise sequences. Both observed series are $I(1)$, but

$$y_t - z_t = u_t - v_t \sim I(0).$$

Hence they are cointegrated with slope $b^\star = 1$. The regression captures a genuine long-run relation. The mean estimate moves from 0.9101 at $T = 100$ to 0.9895 at $T = 1000$, its standard deviation falls from 0.0728 to 0.0093, and mean R^2 rises from 0.8307 to 0.9791. These patterns are consistent with superconsistency, $\widehat{b} - 1 = O_p(T^{-1})$, and $R^2 \to 1$.

We reject $b = 0$ in every simulated sample, but that null is false. This is high empirical power, not evidence of spurious significance. To examine inference, the script also stores `t_true2`, which tests the true $b = 1$ in this case. Its rejection rate is 54.26% at $T = 500$ and 54.74% at $T = 1000$, far above 5%.

The reason is that cointegration does not by itself validate the conventional t distribution. Here the cointegrating error is $e_t = u_t - v_t$, so $\mathrm{Cov}(z_t, e_t) = -1$: the regressor contains the same transitory noise that enters the error with the opposite sign. This dependence produces a nonstandard, shifted distribution for the conventional statistic, visible in the last row of Figure 2. The slope remains consistent, but ordinary t critical values give unreliable inference about its true value. The issue in this design is not serial correlation of e_t, which is itself white noise.

Graphical comparison and interpretation

Figure 2 compares the empirical distributions at $T = 500$. The code uses `tiledlayout(4,3)`, `histogram` with density normalization, and `xline` for the reference lines; `exportgraphics` saves the original figure reproduced below.

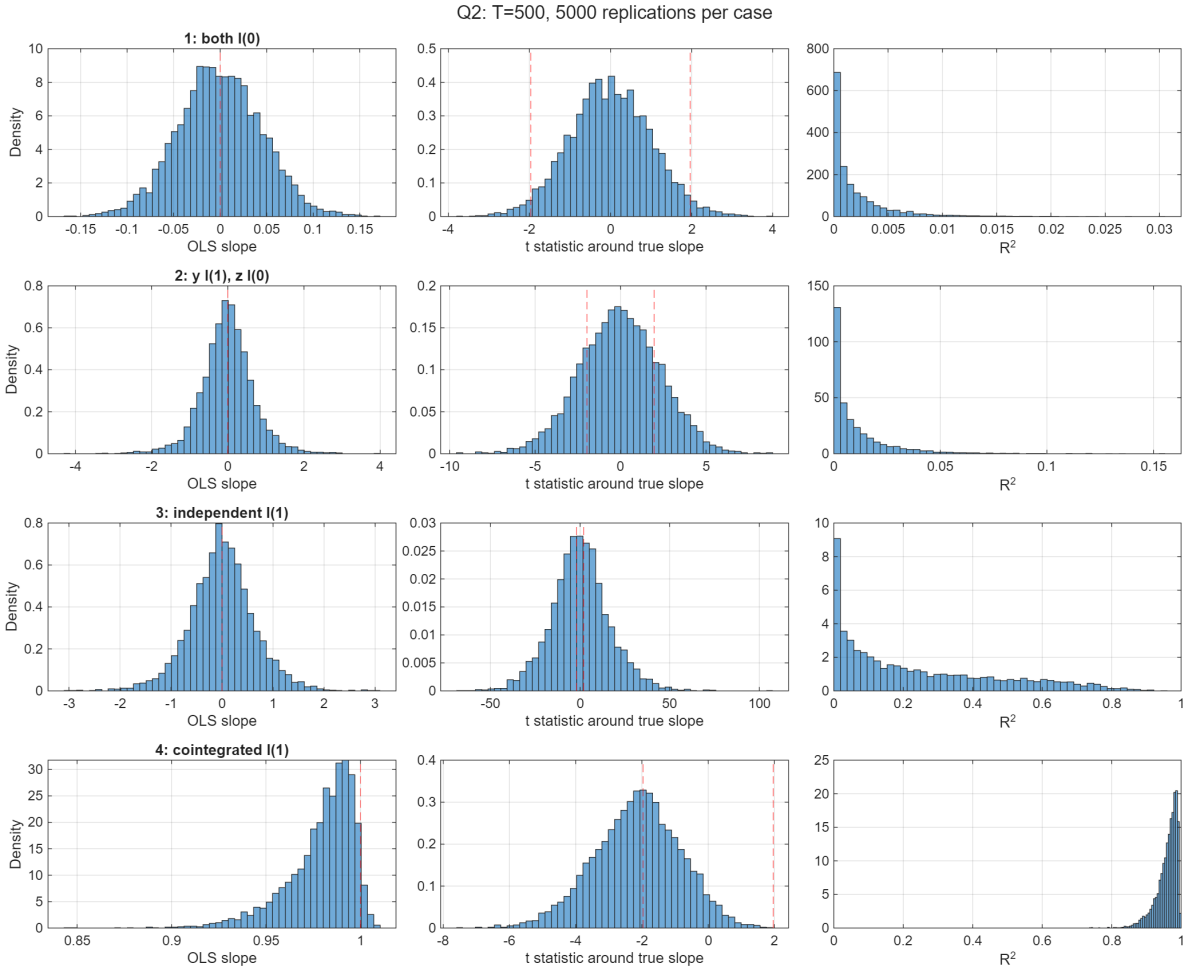


Figure 2: Empirical distributions at $T = 500$, with 5,000 replications per case. Rows correspond to cases 1–4. The left column shows the OLS slope, with a red line at $b^\star$; the middle column shows $t(b^\star)$, with red lines at the conventional two-sided 5% critical values; the right column shows R^2. Thus the final row tests $b = 1$, whereas the first three rows test $b = 0$. Axis scales differ across panels, and the first two R^2 panels are zoomed near zero. Numerical dispersions should therefore be compared using Table 5 and the reported statistics, rather than visual width alone.

The four cases separate two questions: whether the levels regression captures a relationship and whether conventional inference is valid. Independent stationary white noise gives the expected near-zero slope, nominal rejection frequency, and vanishing R^2. With different integration orders, a low R^2 can coexist with seriously misleading tests. With independent random walks, neither a significant coefficient nor a large R^2 establishes a relationship, and the distortion worsens as T increases. Cointegration restores a meaningful long-run coefficient, but inference still needs to account for the properties of the cointegrating regression.

Question 3: Invertibility and Impulse Responses

Suppose that the DGP is

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} 1 & L^2 \\ \dfrac{\beta}{1-\beta} & \dfrac{\beta^2}{1-\beta} + \beta L \end{pmatrix} \begin{pmatrix} \eta_t \\ \varepsilon_t \end{pmatrix},$$

where the shocks are uncorrelated,

$$u_t = \begin{pmatrix} \eta_t \\ \varepsilon_t \end{pmatrix}, \qquad \mathrm{Var}(u_t) = \begin{pmatrix} 1 & 0 \\ 0 & 0.8 \end{pmatrix},$$

and $0 < \beta < 1$*, with* $\beta = 0.6$*. Generate 500 observations for* $(x_t, y_t)'$ *using the DGP, estimate a VAR with four lags and compute impulse responses using a Cholesky identification. Repeat the simulation* N *times and store the impulse responses. Compare the true impulse responses with the estimated ones, using for example the mean and the 2.5th and 97.5th percentiles of the empirical distribution. Are they similar? What is wrong with the DGP? Discuss.*

With $\beta = 0.6$, the DGP can be written explicitly as

$$x_t = \eta_t + \varepsilon_{t-2},$$

and

$$y_t = \frac{\beta}{1-\beta}\eta_t + \frac{\beta^2}{1-\beta}\varepsilon_t + \beta\varepsilon_{t-1}.$$

True Structural Impulse Responses

Let $u_t = (\eta_t, \varepsilon_t)'$ and write the structural moving-average representation as

$$w_t = C(L)u_t = \left(C_0 + C_1 L + C_2 L^2\right) u_t, \qquad w_t = (x_t, y_t)'.$$

The three non-zero coefficient matrices are

$$C_0 = \begin{pmatrix} 1 & 0 \\ \dfrac{\beta}{1-\beta} & \dfrac{\beta^2}{1-\beta} \end{pmatrix}, \quad C_1 = \begin{pmatrix} 0 & 0 \\ 0 & \beta \end{pmatrix}, \quad C_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

We report responses to one-standard-deviation shocks. Since $\mathrm{Var}(u_t) = \mathrm{diag}(1, 0.8)$, the shock-scale matrix is

$$S = \mathrm{diag}(1, \sqrt{0.8}),$$

and the true structural impulse response at horizon h is $\Theta_h = C_h S$. With $\beta = 0.6$, this gives

$$\Theta_0 = \begin{pmatrix} 1 & 0 \\ 1.5 & 0.80498 \end{pmatrix}, \qquad \Theta_1 = \begin{pmatrix} 0 & 0 \\ 0 & 0.53666 \end{pmatrix}, \qquad \Theta_2 = \begin{pmatrix} 0 & 0.89443 \\ 0 & 0 \end{pmatrix}.$$

All true responses are zero from horizon three onwards.

VAR(4) Estimation and Cholesky Identification

For each of $N = 1{,}000$ independent samples of length $T = 500$, we estimate the reduced-form VAR(4)

$$w_t = c + A_1 w_{t-1} + A_2 w_{t-2} + A_3 w_{t-3} + A_4 w_{t-4} + e_t.$$

Let $\widehat{\Sigma}_e$ denote the estimated residual covariance matrix and let P be its lower-triangular Cholesky factor,

$$PP' = \widehat{\Sigma}_e.$$

The ordering $[x, y]$ imposes a zero contemporaneous response of x to the second recursive shock. This restriction is compatible with the zero in the upper-right entry of C_0. The estimated Cholesky shocks have unit variance, so the columns of P already represent one-standard-deviation innovations.

The instructions also describe the equivalent recursive system

$$\begin{aligned} x_t &= b_1 x_{t-1} + c_1 y_{t-1} + \cdots + \eta_t, \\ y_t &= a_2 x_t + b_2 x_{t-1} + c_2 y_{t-1} + \cdots + \varepsilon_t. \end{aligned}$$

The MATLAB script estimates these two equations in the first replication and checks their equivalence with reduced-form OLS followed by Cholesky factorization. The numerical discrepancy is 2.26×10^{-15} in our run, which is only floating-point rounding error.

To compute the responses, we collect the four lag matrices in the companion matrix $\mathcal{A}$ and define

$$\mathcal{D} = \begin{pmatrix} P \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

If $J = (I_2, 0, 0, 0)$ selects the current values of $(x_t, y_t)'$, the estimated impulse response at horizon h is

$$\widehat{\Theta}_h = J\mathcal{A}^h\mathcal{D}, \qquad h = 0, \ldots, 20.$$

Monte Carlo Distribution of the Estimated IRFs

For every response–shock pair and every horizon, we store the 1,000 estimated responses. Figure 3 compares their mean and their 2.5th–97.5th sampling percentiles with the analytical responses.

The discrepancies are systematic rather than small Monte Carlo fluctuations. For example, the true response of x to the second shock is zero at horizon one and 0.8944 at horizon two, whereas the estimated means are 0.3846 and 0.5339. The estimated impact response of y to the first shock is 1.1965 instead of 1.5, and its impact response to the second shock is 1.2810 instead of 0.8050. At later horizons, the small oscillating responses partly reflect finite-sample uncertainty and the approximation of the dynamics by a finite-order VAR.

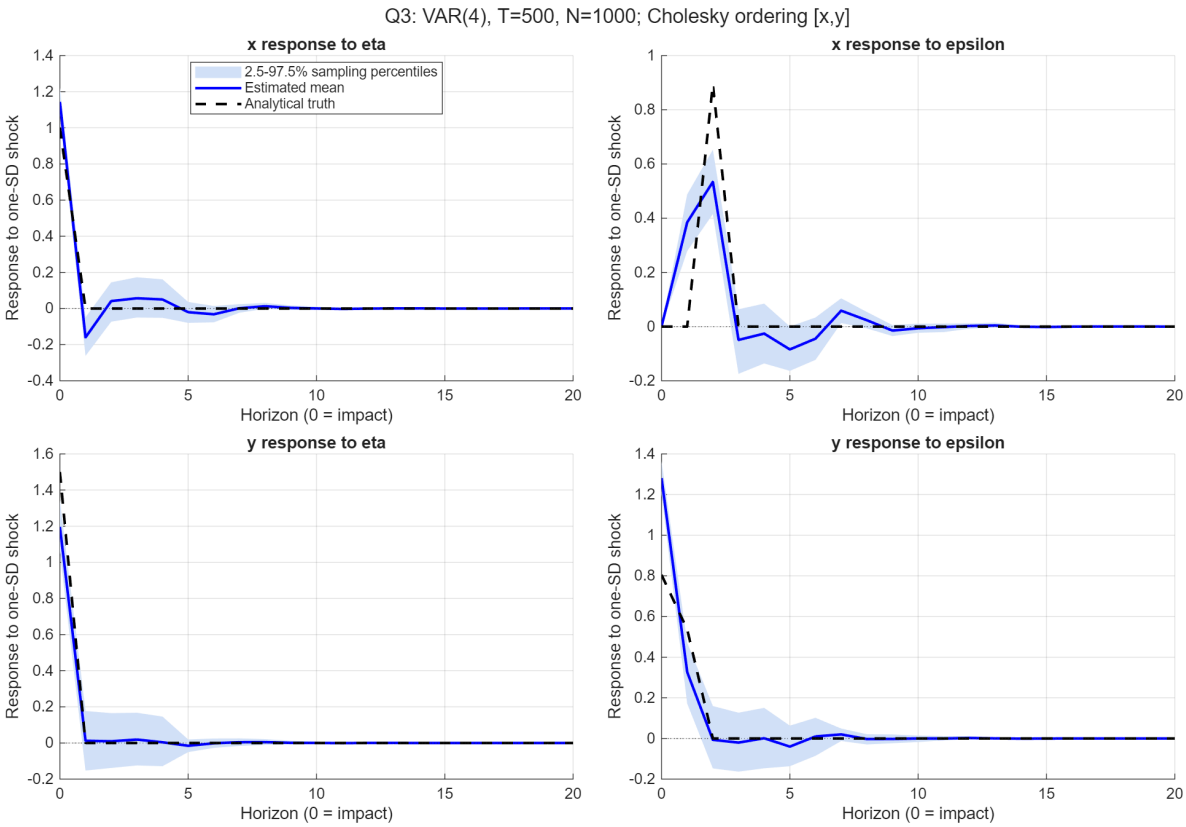


Figure 3: True and estimated impulse responses for the non-fundamental DGP. The solid blue line is the Monte Carlo mean, the shaded region contains the pointwise 2.5th–97.5th percentiles, and the dashed black line is the analytical response. All responses are normalized to one-standard-deviation shocks.

Table 6: Selected true and estimated impulse responses.

Response	Shock	h	True	Mean	2.5%	97.5%
x	η	0	1.0000	1.1439	1.0727	1.2217
x	η	1	0	-0.1599	-0.2621	-0.0487
x	ε	1	0	0.3846	0.2787	0.4876
x	ε	2	0.8944	0.5339	0.4154	0.6523
y	η	0	1.5000	1.1965	1.0596	1.3443
y	ε	0	0.8050	1.2810	1.2023	1.3642
y	ε	1	0.5367	0.3267	0.1723	0.4712

Why the Estimated and True IRFs Differ

The structural MA matrix is

$$C(z) = \begin{pmatrix} 1 & z^2 \\ \dfrac{\beta}{1-\beta} & \dfrac{\beta^2}{1-\beta} + \beta z \end{pmatrix}.$$

Its determinant factors as

$$\begin{aligned} \det C(z) &= \frac{\beta^2}{1-\beta} + \beta z - \frac{\beta}{1-\beta} z^2 \\ &= -\frac{\beta}{1-\beta}(z-1)(z+\beta). \end{aligned}$$

The roots are therefore $z = 1$ and $z = -\beta = -0.6$. The root -0.6 lies strictly inside the unit circle, so the structural MA representation does not have a stable one-sided inverse. The root at one also lies on the boundary and prevents an absolutely summable inverse.

Consequently, the original shocks η_t and ε_t cannot be recovered from current and past observations of $(x_t, y_t)'$. Cholesky identification recovers orthogonal innovations from the fundamental Wold representation of the observables, not the non-fundamental structural shocks used to generate the data. The contemporaneous triangular restriction can be correct and yet remain insufficient to recover the original shocks. This failure of fundamentalness explains why the estimated and true IRFs do not coincide even when the sample size is large.

In a nutshell, the Cholesky ordering is compatible with the DGP on impact, but the DGP is non-invertible because $\det C(z)$ has the root -0.6 inside the unit circle. The VAR therefore identifies fundamental forecast innovations rather than the original structural shocks, and its impulse responses differ systematically from the true structural responses.

Question 4: Granger Causality

> *Read "A New Measure of Monetary Shocks: Derivation and Implications" by Christina D. Romer and David H. Romer (2004). Using the supplied data on US inflation, unemployment, the federal funds rate, and the Romer–Romer monetary policy shocks from 1969Q1 to 1996Q4, estimate a VAR with four lags and test for Granger causality. Do the Romer–Romer shocks Granger-cause the other variables? Do the other variables Granger-cause the Romer–Romer shocks? Discuss.*

Let

$$z_t = \begin{pmatrix} \pi_t \\ u_t \\ i_t \\ RR_t \end{pmatrix},$$

where π_t denotes inflation, u_t unemployment, i_t the federal funds rate, and RR_t the Romer–Romer monetary policy shock.

We estimate a VAR(4),

$$z_t = c + A_1 z_{t-1} + A_2 z_{t-2} + A_3 z_{t-3} + A_4 z_{t-4} + \varepsilon_t.$$

The workbook contains 112 quarterly observations from 1969Q1 to 1996Q4. Four lags leave 108 observations for estimation. Each equation contains an intercept and 16 lag coefficients, so each unrestricted equation has 91 residual degrees of freedom.

For a test with q zero restrictions, we compare the unrestricted and restricted residual sums of squares using

$$F = \frac{(SSR_R - SSR_U)/q}{SSR_U/91}.$$

All tests use the same four-variable VAR. Hence each individual direction is tested conditional on the lags of all the remaining variables.

Do Romer–Romer Shocks Granger-Cause the Macro Variables?

For each macroeconomic variable, the null hypothesis is that all four lag coefficients associated with the Romer–Romer shock are jointly zero.

For example, in the inflation equation,

$$H_0 : A_{1,\pi RR} = A_{2,\pi RR} = A_{3,\pi RR} = A_{4,\pi RR} = 0.$$

The null contains four restrictions in each equation. At the 5% level, we do not reject the absence of predictive content for inflation or unemployment: their p-values are 0.0835 and 0.0765. Both results are nevertheless significant at the 10% level. For the federal funds rate, the statistic is $F = 9.0610$ with $p = 3.31 \times 10^{-6}$, so the null is decisively rejected.

Do the Macro Variables Granger-Cause Romer–Romer Shocks?

We then reverse the direction of causality and test whether lagged inflation, unemployment, and the federal funds rate help predict RR_t.

The individual p-values are 0.6738 for inflation, 0.4856 for unemployment and 0.2708 for the federal funds rate. We therefore do not reject any of the three individual null hypotheses. A final test imposes all 12 macroeconomic lag restrictions jointly. This test gives $F = 0.8708$ and $p = 0.5788$, so the joint null is also not rejected.

Summary of Granger-Causality Results

The results show that lagged Romer–Romer shocks contain strong incremental information for the federal funds rate and weaker evidence for inflation and unemployment. The reverse tests provide no evidence that these three macro variables predict the Romer–Romer series once the shock's own lags and the other variables are included.

This interpretation must remain predictive rather than causal. Granger causality asks whether lagged information improves a forecast; it does not by itself establish structural causation. Romer and Romer construct their measure by removing the component of intended policy changes associated with the Federal Reserve's internal forecasts. The absence of reverse Granger causality is consistent with that intended forecast purging, but it does not prove that the shock measure is structurally exogenous with respect to every possible information set.

Table 7: Conditional Granger-causality tests from the VAR(4).

Cause	Effect	q	F	p-value	Reject 5%	Reject 10%
RR	Inflation	4	2.1295	0.0835	No	Yes
RR	Unemployment	4	2.1878	0.0765	No	Yes
RR	FFR	4	9.0610	3.31×10^{-6}	Yes	Yes
Inflation	RR	4	0.5857	0.6738	No	No
Unemployment	RR	4	0.8694	0.4856	No	No
FFR	RR	4	1.3138	0.2708	No	No
Macro variables jointly	RR	12	0.8708	0.5788	No	No

To conclude, at the 5% level, Romer–Romer shocks Granger-cause the federal funds rate but not inflation or unemployment; the latter two results are significant only at 10%. Inflation, unemployment and the federal funds rate do not Granger-cause the Romer–Romer shocks, either individually or jointly.